

SAI SHRUTHI KULKARNI

Software Developer

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Summary

- Over 3+ years of experience in backend development using Java, Spring Boot, and Python, and frontend technologies like Angular and ReactJS.
- Skilled in designing RESTful APIs and implementing micro-services architecture to enhance scalability and system communication.
- Proficient in deploying applications on cloud platforms with Kubernetes, Docker, and Microsoft Azure, ensuring reliable and seamless deployments.
- Expertise in CI/CD pipelines using Jenkins and conducting testing with JUnit, Mockito, and automated tools for reliable software delivery.
- Experienced in SQL query optimization, developing stored procedures, and working with MySQL databases to improve data retrieval and reporting efficiency.
- Expertise in NoSQL databases like MongoDB and DynamoDB, optimizing database performance for large-scale, high-throughput applications.

Technical Skills

- **Programming & Frameworks:** Java, Spring Boot, TypeScript, Python, Angular, ReactJS, Apache Kafka
- **Cloud & Containerization:** Kubernetes, Docker, Microsoft Azure
- **API Development & Integration:** RESTful APIs, Microservices Architecture, Apache Tomcat
- **DevOps & CI/CD:** Jenkins, CI/CD Pipelines
- **Database Management:** SQL, MySQL, NoSQL, MongoDB, DynamoDB, Stored Procedures, Query Optimization
- **Testing & Quality Assurance:** JUnit, Mockito, Automated Testing Tools, Comprehensive Test Plans
- **Tools & IDEs:** IntelliJ IDEA, Visual Studio, Eclipse
- **Frontend Development:** Angular, ReactJS, Bootstrap, UI/UX Design
- **Scalability & Optimization:** High-throughput systems, Real-time Data Processing
- **Software Development Practices:** SOLID Principles, Modular Architecture, Debugging

Professional Experience

Software Developer | BNY Mellon - Ohio, USA

Oct 2024 - Present

- Developed RESTful APIs using Java with Spring Boot, ensuring high availability and scalability for applications handling over 10,000 monthly active users.
- Conceptualized Apache Kafka for real-time data processing and event streaming, enabling the system to handle high-throughput transactions.
- Designed and deployed microservices using Kubernetes, ensuring seamless orchestration and scaling of applications across Microsoft Azure cloud infrastructure.
- Optimized Apache Tomcat configurations to enhance performance, reducing application load times and improving the overall user experience.
- Led the integration of CI/CD pipelines with Jenkins, improving deployment efficiency and reducing errors.
- Implemented strategies to identify and resolve complex bugs in both the front-end (React) and backend (Java) layers, enhancing system stability.
- Coordinated architecture decisions and software design, ensuring modularity and maintainability of the application by adhering to SOLID principles
- Executed comprehensive test plans and test cases to ensure software quality and compliance with specifications.
- Conceptualized automated testing tools to enhance the QA process, resulting in a 25% reduction in testing time and increased defect detection rates.

Software Developer | Cognizant Technologies — Hyderabad, India

Nov 2021 - Aug 2022

- Configured scalable cloud infrastructure, supporting 10,000 transactions per minute and cutting operational overhead by \$15,000 annually through improved efficiency and automation.
- Engaged in the design and development of sophisticated software solutions using Spring Boot, supporting the execution of over 100,000 API calls daily and ensuring high system throughput with enhanced reliability.
- Implemented targeted bug fixes in TypeScript, improving system performance and streamlining the iterative development process by resolving issues.
- Resolved critical issues, reducing defect rates to 3 incidents per sprint over 3 months. Strengthened code quality and reliability using Python by verifying the functionality of 50+ modules, enabling seamless performance in production environments.
- Streamlined front-end features using Angular and ReactJS, improving user interface responsiveness and reducing application load times, leading to a more efficient and user-friendly experience.
- Leveraged Docker and Kubernetes for containerization and orchestration, enabling seamless deployment across environments and maintaining high application reliability during production runs.

Software Developer | Adani — India

Jul 2020 - Oct 2021

- Formulated a Microservices Architecture for a backend system, creating 3-5 services that improved flexibility and scalability.
- Performed over 500,000 CRUD operations daily on MongoDB for real-time data management, utilizing MongoDB shell, Compass, and Java/Node.js drivers for seamless data access and modification.
- Leveraged Docker to containerize and deploy applications across 3 environments, enhancing the CI/CD pipeline and reducing manual deployment errors.
- Implemented DynamoDB Streams for real-time data processing, triggering AWS Lambda functions for more than 200 data events per minute.
- Collaborated on Front-End Services, including UI/UX development using frameworks like Angular, and Bootstrap.
- Performed scalable and modular backend systems using Spring Boot and microservices architecture.
- Employed Visual Studio and IntelliJ IDEA for the development of software modules, adhering to industry best practices for clean and maintainable code.
- Conducted unit and integration tests for 5+ applications using JUnit and Mockito, ensuring reliable and error-free application performance.
- Diagnosed and resolved 20+ application issues by reviewing logs and performing debugging within Eclipse and Visual Studio.
- Wrote and streamlined SQL queries, stored procedures, and views for core business applications, ensuring data consistency and supporting operational needs.

Frontend Developer Intern | Trigent Software, India

Jan 2020– Jun 2020

- Developed interactive user interfaces using ReactJS, enhancing user experience and reducing page load times across 10+ web applications.
- Built reusable TypeScript components for dynamic data rendering, improving frontend development efficiency by 20%.
- Collaborated with backend teams to integrate RESTful APIs, ensuring seamless data flow and consistency across modules.
- Conducted UI/UX testing to identify usability issues and implemented design improvements based on feedback from user research within 4 weeks.
- Configured Git workflows for version control and streamlined the code deployment process, ensuring efficient collaboration within the team.
- Assisted in migrating legacy code to modern frameworks, completing the migration of 5+ modules in 3 months.

Education

Campbellsville University, Louisville, Kentucky
Master of Science in Computer Science

Aug 2022 - Jun 2024

Guru Nanak University, India
Bachelor of Engineering in Electronics and Communications

Jul 2017 - May 2021